

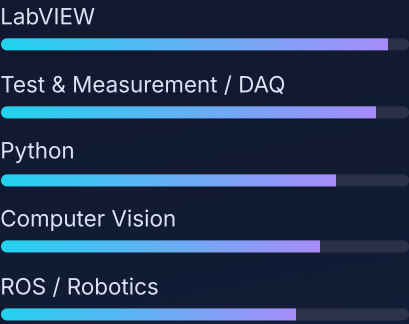
KIRAN SHIVAKUMAR

Application Engineer · LabVIEW · Robotics

CONTACT

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CORE SKILLS



AI & VISION

- OpenCV
- TensorFlow
- MediaPipe
- YOLO v5/v8/v11/26
- DeepSORT

ROBOTICS

- ROS
- Gazebo
- RViz
- SLAM

COMMUNICATION

- RS-485
- RS-232
- Modbus
- CAN Bus
- HART
- TCP/IP
- GPIO
- SCPI
- Telnet
- FTP
- USB

EDUCATION

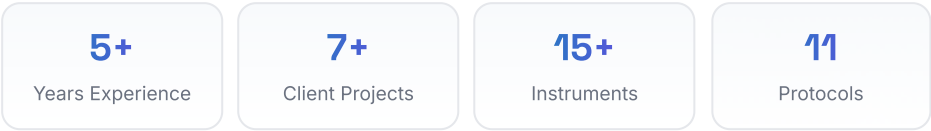
B.E. Mechatronics
The Oxford College of Engineering
2017 – 2021 · CGPA 8.1

LANGUAGES

- English
- Hindi
- Kannada

PROFILE

Application Engineer with **5+ years** of experience designing, developing and deploying **LabVIEW-based test & measurement systems** for aerospace, nuclear, space, power and consumer electronics clients. Hands-on with instrument automation, multi-sensor data acquisition and industrial protocols, and uses **Python, ROS and computer vision** for robotics and intelligent monitoring. Looking to bring this LabVIEW experience to innovative projects.



EXPERIENCE

Application Engineer Feb 2024 – Present
Ducom Aerospace Pvt Ltd · Bengaluru

- Develop **LabVIEW-based test & measurement software** for tribology testing systems.
- Acquire and process data from **force, temperature, humidity, vibration and acoustic emission** sensors.
- Build Python computer vision pipelines for image annotation, segmentation, facial detection and facial landmarks to **track human movement for surveillance**.
- Develop robot movement mapping and **live video visualisation** for a land robotic platform.

Tech: LabVIEW, Python, OpenCV, TensorFlow, MediaPipe, YOLO, DeepSORT, ROS, Gazebo, RViz, SLAM

LabVIEW Developer Sep 2021 – Jan 2024
Access Automation Pvt Ltd · Bengaluru

- Delivered **7 end-to-end automated test & DAQ systems** for clients including **ISRO (SAC), NPCIL, Tata Power, LG India, GACL, Electrohms** and **Oman Fiber Optics**.
- Integrated instruments from Keysight, Yokogawa, HIOKI, Anritsu, EXFO, Advantech and Motwani, plus Mitsubishi FX5U PLCs, over SCPI, GPIB, Modbus and TCP/IP.
- Built automated reporting and remote monitoring dashboards for test data analysis.
- Suggested improvements that made delivered systems run better and fed into new product features.

Tech: LabVIEW, ATE, DAQ, SCPI, GPIB, Modbus, RS-485, PLC integration, Report Generation

KEY PROJECTS

DAQ & Reporting Software for Antenna Testing SAC · ISRO
Configures antennas (frequency, bandwidth, measurement specs), controls DUT/feed positioners and measures radiation patterns.
HW: Keysight MXG N5183B, PNA-X N5264B, Power Meter, Spectrum Analyzer E4446A, positioners

Transformer Testing Bench & Software TATA POWER
Automated bench for no-load, full-load, temperature rise, vector group, magnetising current, magnetic balance, DVDF, HV, turns ratio, winding & insulation resistance tests.
HW: Yokogawa WT500, Motwani XTRM-3A / XWRM-25A / 5 kV IR Tester, FX5U PLC

Tamil

OPERATING SYSTEMS

Windows

Linux

KIRAN SHIVAKUMAR

INSTRUMENTS

- Keysight PNA-X
- Keysight MXG
- Keysight 33210A AWG
- Keysight DSO-X 2012A
- Spectrum Analyzer E4446A
- Yokogawa WT500
- Yokogawa GM10
- HIOKI PW3335
- Anritsu MT9085
- EXFO FTB-500
- Advantech ADAM-4050/4060
- Motwani Test Meters
- Mitsubishi FX5U PLC

DOMAINS

- Aerospace
- Space / RF
- Nuclear Safety
- Power & Energy
- Fibre Optics
- Consumer Electronics
- Tribology
- Surveillance

KEY PROJECTS (CONTD.)

Perimeter Automated Detection System NPCIL · KAKRAPAR APP

Monitors and acknowledges alarms from microwave detectors and PIR sensors across the plant perimeter detection area.
HW: Advantech ADAM-4050, ADAM-4060

Power Consumption Test Automation LG INDIA

Automated monitoring, recording and display of test parameters from power analyzer, proximity sensor and PLC; data used for analysis and report generation.
HW: HIOKI PW3335, Mitsubishi FX5U PLC

Cell Voltage Monitoring System GACL INDIA

Networked DAQ interfacing **19 Yokogawa GM10** units for continuous voltage acquisition and remote monitoring.
HW: Yokogawa GM10 × 19

ATE Software for Frequency Response ELECTROHMS INDIA

Controls AWG and oscilloscope to generate optimal input signals, measure RMS values and compute frequency response in dB (20-log Gain).
HW: Keysight 33210A AWG, Keysight DSO-X 2012A

ConnectOTDR OMAN FIBER OPTICS

Optical fibre test software measuring wavelength, resolution, pulse width and distance.
HW: EXFO FTB-500, Anritsu MT9085

OPEN-SOURCE PROJECTS

- LabVIEW ↔ Python AI Bridge**
Streams LabVIEW DAQ data into a Python ML backend for real-time predictive maintenance.
github.com/kiran10299/LabVIEW-Python-AI-Bridge

AI Hand-Gesture Controller
Webcam mouse control, clicking and scrolling with MediaPipe hand tracking and OpenCV HUD.
github.com/kiran10299/AI-Hand-Gesture-Controller
- Desktop Super Tools**
Watchdog-based auto file organiser and global-hotkey PyQt5 notepad.
github.com/kiran10299/Desktop-Super-Tools

Straw-Hat Reminders
Procedurally animated PyQt5 desktop companion for health reminders.
github.com/kiran10299/Straw-Hat-Reminders